

Each question is worth 5 points for a total of 70 points. The Extra Credit can get you an additional 5 points, making the highest possible score 75 out of 70. Write solutions using COMPLETE sentences.

Problem 1

Each of the following statements can be recast in the if-then form. Please rewrite each of the following sentences in the form "If A, then B".

1. The product of an odd integer and an even integer is even.
2. The square of an odd integer is odd.
3. The square of a prime number is not prime.
4. The sum of three consecutive integers is divisible by three.

Problem 2

Prove that the sum of an odd integer and an even integer is odd.

Problem 3

Suppose a , b , and c are integers. Prove that if $a|b$ and $a|c$, then $a|(b + c)$.

Problem 4

Prove: $(x \wedge y) \vee (x \wedge \neg y)$ is logically equivalent to x .

Hint: use a truth table.

Problem 5

A *bit string* is a list of 0s and 1s. How many length- k bit strings can be made?

Problem 6

There are six different French books, eight different Russian books, and five different Spanish books. In how many different ways can these books be arranged on a bookshelf if all books in the same language are grouped together?

Problem 7

True or False: Please label each of the following sentences as either *true* or *false*. (You do not need to prove your assertions.)

a) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, x + y = 0$.

b) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, x + y = 0$.

c) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, xy = 0$.

d) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, xy = 0$.

e) $\forall x \in \mathbb{N}, \exists y \in \mathbb{N}, x \leq y$.

f) $\exists x \in \mathbb{N}, \forall y \in \mathbb{N}, x \leq y$.

Problem 8

Suppose A and B are finite sets. Given that $|A| = 13$, $|A \cup B| = 26$, and $|A \cap B| = 3$. Determine $|B|$.

Problem 9

For each of the following relations on the set $\{1, 2, 3, 4, 5\}$, determine whether the relation is reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.

a) $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5)\}$

b) $R = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$

c) $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5)\}$

d) $R = \{(1, 1), (1, 2), (2, 1), (3, 4), (4, 3)\}$

e) $R = \{1, 2, 3, 4, 5\} \times \{1, 2, 3, 4, 5\}$

Problem 10

a) Prove that if x and y are both *odd*, then $x \equiv y \pmod{2}$.

b) Prove that if x and y are both even, then $x \equiv y \pmod{2}$.

Problem 11

Let $R = \{(x, y) : x, y \in \mathbb{Z} \text{ and } |x| = |y|\}$

a) Prove that R is an equivalence relation on the integers.

b) Find the equivalence classes $[5]$, $[-2]$, and $[0]$.

Problem 12

The expression $(x + 2)^{50}$ is expanded. What is the coefficient of x^{17} ?

Problem 13

Prove by contradiction: If the sum of two primes is prime, then one of the primes must be 2. You may assume that every integer is either even or odd, but never both.

Problem 14

Prove $n! \leq n^n$ for all positive integer n . You may use any proof technique you prefer.

Extra Credit:

Prove that for all natural numbers n , then

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$